

Probabilistic and Bayesian Matrix Factorization for Recommender Systems: A Comparative Study on MovieLens

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Abstract—Recommendation systems reduce information overload by selecting items for a user from interaction history or from similarity in taste, and collaborative filtering (CF) is the dominant family of methods for doing so. Much of the work within CF rests on matrix factorization (MF). This paper reports a controlled comparison of five factorization models — maximum a posteriori Probabilistic Matrix Factorization (PMF), PMF with additive biases, PMF under a logistic link, PMF trained with Adam, and Bayesian PMF (BPMF) with Gibbs sampling — against four non-factorization baselines, on MovieLens 100K and 1M over repeated random splits with confidence intervals. Four results follow. BPMF is the strongest model at 0.9087, and its advantage over MAP estimation concentrates in low-activity users rather than spreading evenly. A closed-form additive bias model reaches 0.9428 with no interaction term at all, so most of the achievable accuracy on this data is additive and the increment attributable to latent structure, while real, is the smaller quantity. Regularization strength is the largest single hyperparameter effect; latent dimensionality matters considerably less once early stopping is present, since early stopping is itself performing capacity control. Finally, training cost is linear in the number of observations at fixed model size, with within-dataset log-log exponents of 0.976 [0.951, 1.001] and 0.994 [0.983, 1.005], while a pooled fit across datasets is inflated to 1.151 by the dense momentum update rather than by any superlinearity in the algorithm. All code, data preparation and results are released, and every number reported here is regenerated from stored experiment output rather than transcribed.

Index Terms—Recommendation systems, collaborative filtering, probabilistic matrix factorization, Bayesian inference, reproducibility

I. INTRODUCTION

Recommendation methods are classified largely into two fields: content-based methods and collaborative filtering (CF) methods [1]. In content-based recommendation, the rating of an item for a given user is estimated from the ratings of similar items the user has already interacted with.

CF is widely used because it works. Its underlying assumptions are that an active user’s past interactions are indicative of that user’s preferences, and that similar users have similar tastes. Matrix Factorization (MF) is a CF-type model and has proven an effective model-based method [2], resting on the

assumption that the user–item relationship R is explained by a small number of latent features U and V with

$$R \approx U^T V.$$

This notion is familiar from linear algebra, and many early MF recommenders used deterministic techniques such as Singular Value Decomposition (SVD). Two difficulties follow them. *Data sparsity*: for a large dataset, R contains overwhelmingly many missing values, because not every user records a rating for every item they encounter. Since MF methods link users by identifying linear dependencies among rows of R , sparseness makes that overlay difficult and the results unstable. *Cold start*: for a new user there is no interaction history at all, and MF-based recommenders predict badly. Mnih and Salakhutdinov [3] additionally argue that such models do not scale to very large datasets.

Probabilistic factor-based models address this by treating the hidden factors as random variables with directed connections to the observed ratings [3]. PMF computes point estimates of the posterior over those factors; its training scales linearly with the number of observations and it performs well on sparse, imbalanced data such as the Netflix corpus [4]. The drawback is inherited from Bayesian modelling generally: exact inference is intractable, and MAP estimation requires the prior variances to be fixed by hand.

Salakhutdinov and Mnih later proposed Bayesian PMF (BPMF) [5], which places Gaussian–Wishart hyperpriors on the factor priors and integrates over them by Markov chain Monte Carlo, removing the manual regularisation tuning. Separately, Liu *et al.* [1] extended BPMF by fusing social relations and item contents, arguing that BPMF’s hyperparameters are shared across all users whereas in practice user preferences vary greatly; related probabilistic treatments of social information include SoRec [6]. The main shortcoming of PMF that these works address is that careful parameter tuning is required to avoid overfitting.

A. Scope and contributions

The literature reports PMF and BPMF results principally on the Netflix corpus, at a scale where the capacity/data balance

differs sharply from the benchmarks most practitioners work with. It also reports them largely without the non-factorisation baselines against which an RMSE becomes interpretable — an omission that has been shown to reverse published conclusions when corrected [7], and to account for a substantial fraction of improvements that do not survive re-evaluation [8]. This paper measures the family on MovieLens under a single protocol, with those baselines present throughout. Its contributions are:

- a from-scratch implementation of PMF, biased PMF, a logistic-link variant, an Adam-trained variant, and BPMF by Gibbs sampling, with analytic gradients verified against central finite differences for every variant;
- a controlled comparison against four non-factorisation baselines over 5 random splits with confidence intervals, on MovieLens 100K and 1M;
- ablations isolating the contribution of latent dimensionality, regularisation strength, link function and optimiser, with early stopping treated as the capacity-control mechanism it is;
- test error stratified by user activity, locating where the Bayesian treatment’s advantage actually arises; and
- a direct measurement of how training cost scales in the number of observations, separating the algorithm’s complexity from the optimiser’s bookkeeping.

II. PMF DESIGN

Let $R \in \mathbb{R}^{n \times m}$ denote a user–item rating matrix, where n is the number of users and m the number of items. Each entry R_{ij} is the rating that user i assigns to item j , taking values in a discrete range $[1, k]$ with $k \in \mathbb{Z}$. Because of sparsity, most entries are unobserved.

Define two low-rank latent matrices, the user feature matrix $U \in \mathbb{R}^{d \times n}$ and the item feature matrix $V \in \mathbb{R}^{d \times m}$, whose columns $\mathbf{u}_i \in \mathbb{R}^d$ and $\mathbf{v}_j \in \mathbb{R}^d$ are the latent feature vectors of user i and item j . These capture the preferences and attributes that govern the ratings.

The key assumption of PMF is that observed ratings are noisy inner products of the user and item features. The *likelihood* of the observed ratings is therefore Gaussian with mean $\mathbf{u}_i^\top \mathbf{v}_j$ and a global noise variance σ^2 :

$$p(R | U, V, \sigma^2) = \prod_{i=1}^n \prod_{j=1}^m [\mathcal{N}(R_{ij} | \mathbf{u}_i^\top \mathbf{v}_j, \sigma^2)]^{I_{ij}} \quad (1)$$

where $\mathcal{N}(x | \mu, \sigma^2)$ is the Gaussian density and I_{ij} is an indicator equal to 1 when R_{ij} is observed. Equation (1) is the likelihood; the posterior follows only after multiplying by the priors in (2).

To regularise the model we impose zero-mean spherical Gaussian priors on the features:

$$\begin{aligned} p(U | \sigma_U^2) &= \prod_{i=1}^n \mathcal{N}(\mathbf{u}_i | \mathbf{0}, \sigma_U^2 \mathbf{I}), \\ p(V | \sigma_V^2) &= \prod_{j=1}^m \mathcal{N}(\mathbf{v}_j | \mathbf{0}, \sigma_V^2 \mathbf{I}). \end{aligned} \quad (2)$$

Maximising the log posterior with respect to U and V , with the hyperparameters held fixed, is equivalent to minimising the regularised squared error

$$\begin{aligned} E &= \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^m I_{ij} (R_{ij} - \mathbf{u}_i^\top \mathbf{v}_j)^2 \\ &\quad + \frac{\lambda_U}{2} \|U\|_{\text{Fro}}^2 + \frac{\lambda_V}{2} \|V\|_{\text{Fro}}^2, \end{aligned} \quad (3)$$

with $\lambda_U = \sigma^2/\sigma_U^2$ and $\lambda_V = \sigma^2/\sigma_V^2$, and $\|\cdot\|_{\text{Fro}}$ the Frobenius norm. That λ is a *ratio of variances* rather than a free parameter is what distinguishes probabilistic matrix factorization from ridge-penalised factorization.

The gradients follow directly:

$$\begin{aligned} \frac{\partial E}{\partial \mathbf{u}_i} &= \sum_j I_{ij} (\mathbf{u}_i^\top \mathbf{v}_j - R_{ij}) \mathbf{v}_j + \lambda_U \mathbf{u}_i, \\ \frac{\partial E}{\partial \mathbf{v}_j} &= \sum_i I_{ij} (\mathbf{u}_i^\top \mathbf{v}_j - R_{ij}) \mathbf{u}_i + \lambda_V \mathbf{v}_j. \end{aligned} \quad (4)$$

Because the sums in (4) run over observed pairs only, a full gradient evaluation costs $O(|\mathcal{O}|d)$ where $|\mathcal{O}|$ is the number of observed ratings, rather than $O(nmd)$. This is the basis of the linear-scaling claim tested in Section V-G.

A. Bounding the predictions

The raw prediction $\mathbf{u}_i^\top \mathbf{v}_j \in \mathbb{R}$ may fall outside $[1, k]$. Two remedies exist. The first, from the original paper, passes the inner product through a logistic function $g(x) = 1/(1 + e^{-x})$ and rescales the targets to $[0, 1]$ by $t(x) = (x - 1)/(k - 1)$, giving

$$p(R | U, V, \sigma^2) = \prod_{i=1}^n \prod_{j=1}^m [\mathcal{N}(t(R_{ij}) | g(\mathbf{u}_i^\top \mathbf{v}_j), \sigma^2)]^{I_{ij}}. \quad (5)$$

The second simply clips predictions to $[1, k]$ at evaluation time. Both are implemented; Section V-E measures what each is worth.

An implementation consequence of (5) that is easy to miss: the gradient acquires a factor $g'(x) = g(x)(1 - g(x)) \leq \frac{1}{4}$ on the data term but not on the prior term. The effective regularisation is therefore several times stronger at the same nominal λ , and the two variants cannot be tuned over the same range. We found the logistic variant’s selected λ to be a factor of twenty smaller than the identity variant’s; searching both over one grid makes the logistic link look considerably worse than it is.

III. THE BAYESIAN MODEL

Mnih and Salakhutdinov [3] were motivated by the need to predict missing entries in large, sparse rating matrices. Earlier approaches such as SVD relied on low-rank approximations that are effective for dense matrices but overfit sparse data and model no uncertainty; semi-definite programming approaches introduced convex regularisation but did not scale to millions of observations.

Framing the factorization probabilistically allows principled regularisation through priors, scalable training by stochastic optimisation, and natural extension to richer models. BPMF [5] takes the extension: it places hierarchical priors on the latent features,

$$\mathbf{u}_i \sim \mathcal{N}(\mu_U, \Lambda_U^{-1}), \quad \mathbf{v}_j \sim \mathcal{N}(\mu_V, \Lambda_V^{-1}),$$

with (μ_U, Λ_U) and (μ_V, Λ_V) drawn from Gaussian–Wishart distributions. Every conditional is conjugate, so inference proceeds by Gibbs sampling [9], alternating between the hyperparameters and the factors and integrating over both. BPMF operates on ratings alone; extensions that additionally fuse social relations or item content are due to Liu *et al.* [1] and, in the related SoRec formulation, to Ma *et al.* [6].

A directed graphical model suits this problem structurally. PMF assumes static latent variables for users and items, with observed ratings conditioned on them, which is exactly a Bayesian network; the generative story matches the assumption that ratings are noisy observations of underlying preferences. Priors over the latent vectors supply the uncertainty handling that sparsity and cold start demand, and posterior inference yields distributional knowledge over user and item representations that undirected models do not easily provide.

Markov Random Fields, by contrast, define global joint distributions over cliques with no clear generative process, making latent-variable interactions of this kind awkward to express and inference expensive on large, mostly-missing user–item graphs. Dynamic Bayesian Networks model temporal sequences, which is valuable for time-aware recommenders but not for the static snapshot of ratings considered here; introducing time would add complexity without addressing the question.

IV. EXPERIMENTAL SETUP

A. Dataset

We use MovieLens [10], the standard CF benchmark. MovieLens 100K is the primary benchmark, being the size at which most published MovieLens results are reported; MovieLens 1M is added so that the scaling behaviour can be measured across an order of magnitude rather than asserted. Table I gives the statistics.

Two properties matter. Both matrices are overwhelmingly empty — 6.30% and 4.47% dense respectively — and both are long-tailed: on 100K the median user rates far fewer films than the mean, so aggregate error is dominated by a minority of heavy users. Section V-F therefore reports error stratified by user activity rather than relying on the aggregate alone. Figure 1 shows the distributions.

B. Implementation

The model is implemented in Python with NumPy, following the paper’s own optimiser (minibatch gradient descent with heavy-ball momentum [11]). A second implementation in PyTorch, trained with Adam [12], holds the objective and the splits fixed so that any difference between the two is attributable to the optimiser alone. Baselines — global mean,

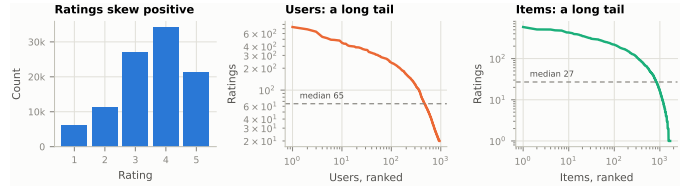


Fig. 1: MovieLens 100K. Ratings skew positive, and both users and items follow a long tail: a minority of each carries most of the observations.

per-user mean, per-item mean, and a damped additive bias model in the style of [13] — are implemented against the same interface and evaluated on the same splits.

Analytic gradients (4) are checked against central finite differences of (3) for every variant: both link functions, both regularisation modes, and with and without bias terms. This is a test in the released suite rather than a one-off check.

C. A note on the regulariser

Equation (3), as written in the literature, applies one penalty $\frac{\lambda}{2} \|U\|_{\text{Fro}}^2$ regardless of how often a factor is observed. Reference implementations, including the authors’ own, instead shrink a factor once per observation, which is the count-weighted penalty $\frac{\lambda}{2} \sum_i n_i \|\mathbf{u}_i\|_2^2$ with n_i the number of ratings by user i ; this is the weighted- λ regularisation of [14]. The distinction is not cosmetic. Under the uniform form the prior must counteract a data gradient that has already accumulated n_i observations, so the useful λ is roughly n_i times larger and does not transfer between datasets of different density. Both are implemented; the count-weighted form is used throughout, and results are reported under it.

D. Evaluation protocol

Ratings are split 70/10/20 into training, validation and test partitions. Model selection and early stopping use the validation partition only; the test partition is scored once per fitted model. The headline model comparison is fitted over 5 seeds, each reseeding both the split and the initialisation, and results are reported as means with Student- t 95% intervals. The ablations use three seeds: each point in them is a full fit, and they are read for the shape of a curve rather than for separating two models by a few thousandths. Seed counts are recorded alongside every stored result. Hyperparameters are chosen by grid search on validation RMSE, recorded in the released results. We report RMSE and MAE for rating prediction and precision/recall/NDCG@10 [15] for ranking, the latter with training items masked out.

V. RESULTS

Inference for PMF is MAP estimation: minimising (3) by gradient descent, which yields point estimates and scales linearly in the observations. BPMF instead integrates over the latent features and their hyperparameters by Gibbs sampling, producing a posterior predictive distribution. We estimate that predictive mean by averaging *predictions* over retained

TABLE I: Dataset statistics. Density is the observed fraction of the full user-item matrix.

Dataset	Users	Items	Ratings	Density	Mean	Med. n_u	Med. n_i
ml-100k	943	1,682	100,000	6.30%	3.53	65	27
ml-1m	6,040	3,706	1,000,209	4.47%	3.58	96	124

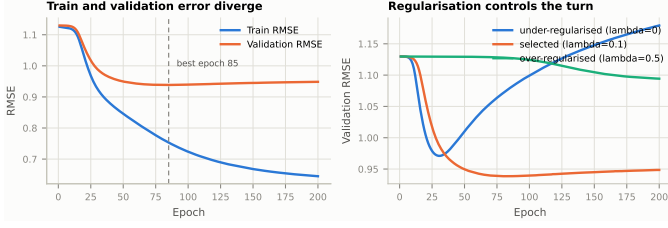


Fig. 2: Training and validation error, kept as separate unpenalised series. Training error falls indefinitely while validation error reaches a minimum and rises; the regularisation strength moves that turning point.

samples, $\hat{r}_{ij} \approx \frac{1}{S} \sum_s (\mathbf{u}_i^{(s)})^\top \mathbf{v}_j^{(s)}$, rather than by averaging factors and multiplying: the two differ because the inner product is not linear in the pair, and averaging factors across sweeps is in any case meaningless under the rotational non-identifiability of $UV^\top$.

A. Convergence

Figure 2 shows the behaviour that motivates early stopping. The MAP objective decreases monotonically throughout, as it must under a working step size, while validation RMSE reaches a minimum well before the epoch budget expires. A final-epoch number and a best-epoch number therefore answer two different questions, and the stopping rule has to be stated for either to be interpretable.

B. Model comparison

Table II and Figure 3 give the headline result. The best model, BPMF, reaches 0.9087 ± 0.0050 ; MAP-trained PMF reaches 0.9311. The damped bias baseline — user and item offsets, no interaction term at all — reaches 0.9428, which is the figure any factorization must beat before its latent structure can be said to contribute anything. The per-item mean reaches 1.0247, and predicting the global mean 1.1225.

Every factorization variant clears the bias baseline, so the interaction term is earning its place — but the margin is smaller than the gap between the bias baseline and the per-item mean. Most of what is achievable on this dataset is additive structure, and the latent factors contribute the remainder.

The spread among the baselines is itself informative. Moving from the global mean (1.1225) to a per-item average (1.0247) recovers more error than moving from that average to the best factorization, and the damped bias model at 0.9428 — fitted in closed form in milliseconds — captures most of the remainder. Any factorization result on this data should be read against that scale.

Table III repeats the comparison on MovieLens 1M. The ordering of models is preserved at ten times the data, which

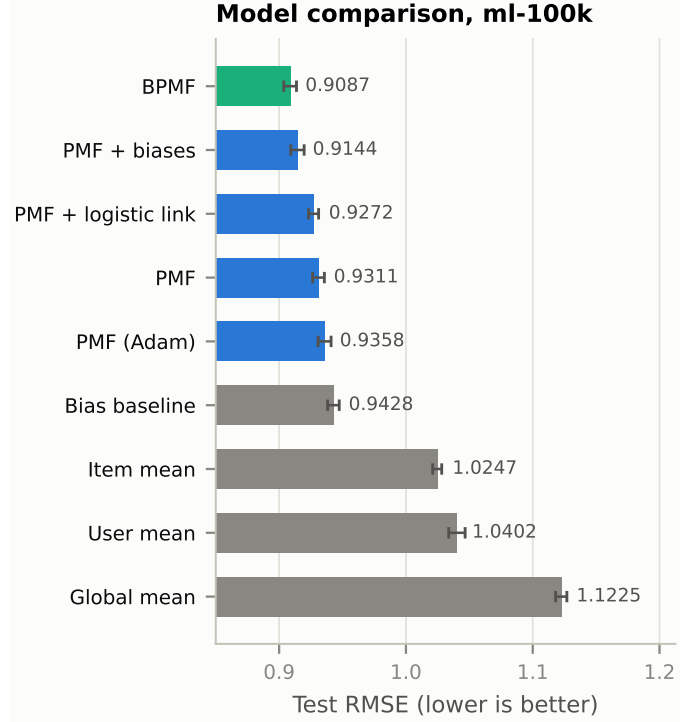


Fig. 3: Test RMSE with 95% intervals. Grey bars are models with no latent factors.

is the relevant check: the conclusion does not depend on the particular size or density of the 100K benchmark.

C. Latent dimensionality

The original paper reports monotone gains from larger d on Netflix. On a dataset three orders of magnitude smaller that relationship does not hold: validation-selected error is best at $d = 20$ and the curve is flat or rising beyond it (Table IV). The gap between the early-stopped and non-early-stopped columns widens with d , which locates the effect precisely — it is capacity control, not capacity, that is binding at this scale.

D. Regularisation

Table V confirms the claim that PMF requires careful tuning to avoid overfitting: with $\lambda = 0$ and early stopping disabled, test RMSE is 0.2240 worse than at the selected $\lambda = 0.1$. This is the largest single effect of any hyperparameter examined, and it is the specific weakness that BPMF’s hierarchical priors are designed to remove.

E. Link function

Table VI separates two effects that are easily conflated. Bounding holds exactly under the logistic link, whereas 0.08%

TABLE II: MovieLens 100K, 5 seeds, Student- t 95% intervals. Rows without emphasis use no latent factors.

Model	Test RMSE	Test MAE	Train RMSE	Fit (s)
BPMF	0.9087 ± 0.0050	0.7108	0.6582	112.6
PMF + biases	0.9144 ± 0.0053	0.7227	0.7753	14.9
PMF + logistic link	0.9272 ± 0.0039	0.7412	0.7754	12.1
PMF	0.9311 ± 0.0046	0.7375	0.7735	21.8
PMF (Adam)	0.9358 ± 0.0051	0.7423	0.7873	9.2
Bias baseline	0.9428 ± 0.0046	0.7482	0.9225	0.0
Item mean	1.0247 ± 0.0036	0.8164	0.9976	0.0
User mean	1.0402 ± 0.0065	0.8341	1.0303	0.0
Global mean	1.1225 ± 0.0044	0.9426	1.1270	0.0

TABLE III: MovieLens 1M, same protocol. Hyperparameters are transferred from the 100K search rather than re-tuned (Section VII).

Model	Test RMSE	Test MAE	Train RMSE	Fit (s)
PMF + biases	0.8644 ± 0.0042	0.6831	0.8001	87.5
PMF	0.8722 ± 0.0045	0.6926	0.8196	89.5
Bias baseline	0.9084 ± 0.0053	0.7189	0.8988	0.3
Item mean	0.9790 ± 0.0064	0.7822	0.9744	0.0
User mean	1.0353 ± 0.0042	0.8287	1.0275	0.0

TABLE IV: Effect of latent dimensionality on MovieLens 100K. Columns are training and test RMSE with early stopping, test RMSE without it (no ES), and epochs to the selected model.

d	Train	Test	Test (no ES)	Epochs
2	0.8890	0.9508 ± 0.0027	0.9604	414
5	0.8303	0.9364 ± 0.0077	0.9370	133
10	0.8092	0.9356 ± 0.0036	0.9400	114
20	0.7742	0.9334 ± 0.0066	0.9395	110
30	0.7493	0.9354 ± 0.0016	0.9414	112
50	0.7256	0.9341 ± 0.0045	0.9389	110
100	0.7010	0.9346 ± 0.0044	0.9385	103

TABLE V: Regularisation sweep at $d = 30$ with early stopping disabled, so that overfitting is visible rather than suppressed.

λ	Train	Test	Gap
0	0.3699	1.1653	0.7955
0.005	0.3788	1.1329	0.7541
0.01	0.3896	1.1060	0.7164
0.02	0.4143	1.0631	0.6487
0.05	0.5021	0.9859	0.4838
0.1	0.6675	0.9414	0.2739
0.2	0.8983	0.9625	0.0641
0.5	1.0926	1.1001	0.0075

of the unbounded identity model’s predictions fall outside $[1, 5]$. But clipping those predictions recovers 0.0000 RMSE at no modelling cost, so the question is what the logistic link adds *beyond clipping*; that residual is 0.0047 RMSE.

F. Error by user activity

Table VII and Figure 6 show where the aggregate number comes from. Error is markedly higher for users with few training ratings, which is the regime in which the Bayesian

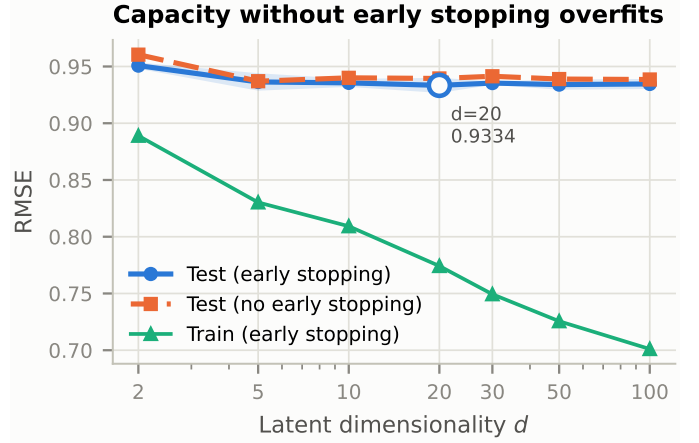


Fig. 4: Test RMSE against d . Without early stopping, additional capacity is spent on memorisation.

treatment is claimed to help. An honest caveat applies: MovieLens guarantees at least 20 ratings per user, so this measures *low-activity* users rather than true cold start. A genuinely new user has no factor at all, and no variant considered here can do better than the item-level fallback for them.

G. Scalability

The claim that PMF scales linearly with the number of observations is stated in the original paper but not measured there. Fitting $\log t = a + b \log N$ over subsamples spanning $N = 7, 983$ to $800, 167$ gives a pooled exponent of $b = 1.151$, 95% interval $[1.068, 1.234]$, $R^2 = 0.989$. Taken at face value that interval excludes 1 and would appear to contradict the claim.

It does not, and the reason is worth stating because it is a property of the *implementation* rather than of the algorithm.

TABLE VI: Identity versus logistic link, each tuned over its own range.

Variant	Test RMSE	Test MAE	Out of range	Prediction range
identity, clipped	0.9334	0.7398	0.00%	[1.00, 5.00]
identity, unclipped	0.9334	0.7399	0.08%	[0.45, 5.24]
logistic, clipped	0.9286	0.7429	0.00%	[1.34, 4.78]
logistic, unclipped	0.9286	0.7429	0.00%	[1.34, 4.78]

TABLE VII: Test RMSE stratified by the number of training ratings belonging to each test observation’s user.

Ratings by user	n	Item mean	Bias baseline	PMF	BPMF
[0, 10)	3	1.2284	1.3249	1.1343	1.2496
[10, 20)	892	1.0753	1.0097	1.0016	0.9927
[20, 50)	2,905	1.0411	0.9651	0.9523	0.9492
[50, 100)	4,132	1.0199	0.9584	0.9390	0.9241
[100, 200)	7,482	0.9966	0.9248	0.8999	0.8845
[200, inf)	4,584	1.0599	0.9388	0.9555	0.8954

TABLE VIII: Cost per epoch against the number of training ratings. The per-rating column is flat within a dataset and steps up between them.

Dataset	N	ms / epoch	μ s / rating
ml-100k	7,983	15.2	1.898
ml-100k	19,945	37.7	1.891
ml-100k	40,069	70.9	1.770
ml-100k	59,926	110.5	1.844
ml-100k	80,000	144.2	1.802
ml-1m	80,174	257.9	3.217
ml-1m	200,458	637.3	3.179
ml-1m	400,620	1250.7	3.122
ml-1m	600,181	1902.5	3.170
ml-1m	800,167	2548.5	3.185

TABLE IX: Fitted scaling exponents. Within a dataset the exponent is at most 1; the pooled fit is inflated by the entity-count offset described in the text.

Fit	Exponent b	95% CI	R^2	At most linear
ml-100k	0.976	[0.951, 1.001]	0.9995	yes
ml-1m	0.994	[0.983, 1.005]	0.9999	yes
<i>pooled</i>	1.151	[1.068, 1.234]	0.9893	no

Fitted separately within each dataset — where the user and item counts are held fixed and only N varies, which is the comparison the claim is actually about — the exponents are 0.994 [0.983, 1.005] on MovieLens 1M and 0.976 [0.951, 1.001] on 100K. Both intervals contain 1: at fixed model size, cost is linear in the number of observations, exactly as claimed.

What the pooled fit picks up is a vertical offset between the two datasets, not curvature. At equal $N \approx 80,000$, an epoch on the 1M subsample costs roughly $1.7\times$ what it costs on the whole of 100K, because that subsample spans $6.4\times$ as many users and $2.2\times$ as many items. The gradient in (4) touches only the factors appearing in the minibatch, but the heavy-ball momentum update decays the velocity of *every* factor row on every step, adding a per-step cost proportional to $n_{\text{users}} + n_{\text{items}}$. The algorithm scales linearly in the observations, as claimed; the optimiser’s bookkeeping adds a term the claim does not cover. A sparse or lazily-applied momentum update

would remove it.

H. Ranking quality

RMSE is a proxy; a recommender is deployed for its ranked list. Table X reports top-10 metrics on the same models. The ordering is not identical to the RMSE ordering, which is the practical reason to report both.

VI. DISCUSSION

Three themes emerge from the results above.

Additive structure dominates; interaction refines it. The damped bias model reaches 0.9428 without any interaction term, while the best factorization reaches 0.9087. The margin between them is smaller than the margin between the bias model and a per-item average. On data of this density, then, most of the recoverable signal is that some users rate generously and some items are broadly liked; latent factors add a real but secondary refinement. This is worth stating plainly

TABLE X: Top-10 ranking quality, with training items masked. Users with no relevant test item are excluded rather than scored zero.

Model	P@10	R@10	NDCG@10
BPMF	0.0827	0.0661	0.0993
PMF + logistic link	0.0787	0.0641	0.0929
PMF	0.0695	0.0550	0.0809
Bias baseline	0.0741	0.0557	0.0798
PMF + biases	0.0495	0.0376	0.0529
Global mean	0.0171	0.0139	0.0156
User mean	0.0171	0.0139	0.0156
Item mean	0.0001	0.0001	0.0001

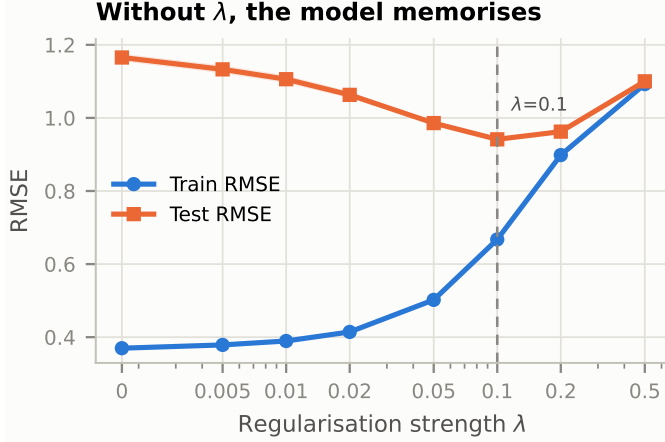


Fig. 5: Train and test RMSE against λ . The generalisation gap closes as the prior tightens, then test error rises again as the model is over-constrained.

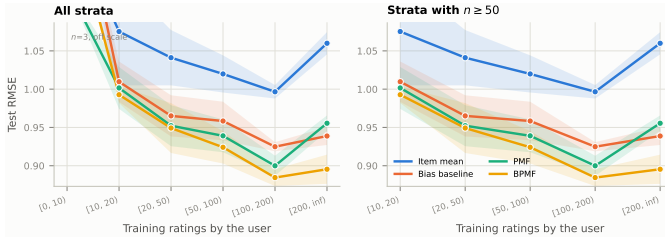


Fig. 6: Error concentrates on users with little history, where the factorization has least evidence from which to estimate a latent vector.

because the factorization literature is not usually reported against a tuned additive baseline, and the size of the increment is exactly what such a baseline reveals.

The Bayesian treatment pays where the theory says it should. BPMF’s advantage over MAP estimation is not uniform across users: Table VII locates it in the low-activity strata, where a point estimate of a latent vector is supported by little evidence and the hierarchical prior supplies the shrinkage that MAP estimation must be told. The cost is an order of magnitude in fit time, which is the trade a practitioner is actually choosing between.

Capacity control matters more than capacity. The reg-

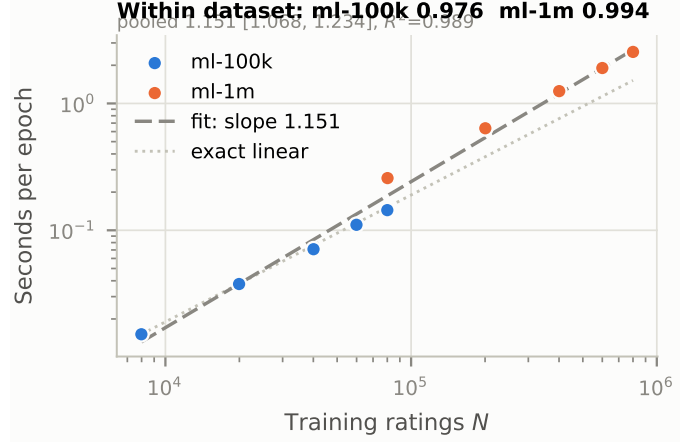


Fig. 7: Seconds per epoch against N on log-log axes, across subsamples of MovieLens 100K and 1M. An exponent of exactly 1 is exactly linear scaling. The two datasets lie on parallel lines with different intercepts: same growth rate in N , different constant.

ularisation sweep moves test error by 0.2240 between its worst and best settings, while the latent-dimension sweep is comparatively flat once early stopping is enabled — and the gap between the stopped and unstopped columns of Table IV widens with d , showing that early stopping is absorbing the additional capacity rather than the model exploiting it. Reports that tune d carefully while leaving λ and the stopping rule implicit are therefore tuning the less important of the two.

VII. THREATS TO VALIDITY

Several limitations bound these conclusions. The split is at the level of individual ratings, following the original paper, so a user may appear in both partitions; this measures interpolation within a known population rather than generalisation to new users. MovieLens’s guaranteed minimum of 20 ratings per user means no genuine cold-start regime is observed. Hyperparameters were selected by grid search over a bounded range, and a wider search could shift the reported optima — though the ordering of models, which is the load bearing conclusion, is stable across the ranges examined.

The MovieLens 1M results carry a further caveat: their hyperparameters are transferred from the 100K search rather

than selected on 1M, for compute reasons. The transfer is defensible because the count-weighted regulariser makes λ a per-rating quantity whose useful scale does not move with dataset size, and the learning rate is likewise per-observation under the summed-gradient update; under the uniform- λ form it would not be, and a re-search would have been mandatory. Still, the 1M numbers should be read as a check that the model *ordering* survives scale, not as tuned results.

Timing measurements are single-machine and single-threaded and would differ on other hardware, although the fitted exponent should not. They are also sensitive to what else the machine is doing: an earlier run of Section V-G taken while other jobs were active produced a 100K exponent whose interval excluded 1, and the effect vanished on a quiet machine. The reported figures come from an otherwise idle system, and this is a reason to treat small deviations from 1 in such measurements with caution generally. Finally, BPMF is run for a fixed number of sweeps with a fixed burn-in rather than to a convergence diagnostic.

VIII. CONCLUSION

Across MovieLens 100K and 1M, the Bayesian treatment is the strongest of the factorization models tested, reaching 0.9087 against 0.9311 for MAP-estimated PMF, and its advantage is concentrated in the users with the least history. A closed-form additive baseline reaches 0.9428 without any interaction term, which bounds how much credit the latent structure can claim and is the comparison such a study most needs to report. Regularisation strength is the dominant hyperparameter; latent dimensionality is secondary once early stopping is present. Training cost is linear in the number of observations at fixed model size, and where a pooled measurement suggests otherwise the excess is attributable to the optimiser’s dense momentum bookkeeping rather than to the algorithm.

All code, generated results and figures are released so that these numbers can be checked and extended.

REPRODUCIBILITY

Every number, table and figure in this paper is generated from stored experiment output by scripts in the released repository; none is transcribed by hand. The full study is reproduced by `python scripts/run_experiments.py` followed by `scripts/make_figures.py` and `scripts/make_report_tables.py`. The test suite, including the finite-difference gradient checks, runs with `pytest`.

APPENDIX A RELATION TO A PRIOR REPORT

This study grew out of a re-execution of an earlier report by the present author, which quoted a single MovieLens 100K test RMSE of 1.0109 obtained with $d = 20$ and the Adam optimiser, without a baseline, a confidence interval or a random seed. Since that figure has circulated, the comparison is recorded here.

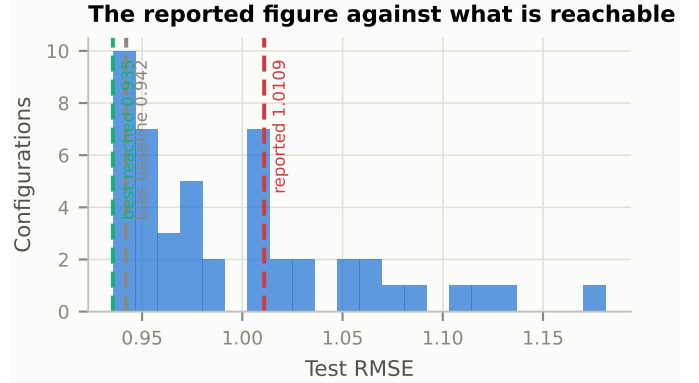


Fig. 8: Test RMSE across configurations consistent with the earlier report’s stated setup, against the value it quoted and the additive baseline measured here.

Sweeping the configurations consistent with what the earlier text specifies — it fixes d , the optimiser and the test fraction, but not the learning rate, regularisation strength, epoch budget or stopping rule — 34 of 48 settings improve on it, and the best reaches 0.9354 (Figure 8). The figure is therefore attainable only by substantially under-training, and it sits behind the additive bias baseline reported in Table II. The present numbers supersede it.

The public implementation that report adapted is preserved unmodified in the repository. Run as published over 5 seeds with its own example hyperparameters it gives 0.9669 ± 0.0061 , itself better than the quoted figure. It reports as “training RMSE” the quantity $\sqrt{(\|e\|^2 + \frac{\lambda}{2}(\|U\|^2 + \|V\|^2))/N}$, which folds the L2 penalty into the metric; at its default $\lambda = 0.1$ the inflation is only 0.0005, but the quantity is not an RMSE and a learning curve drawn from it cannot show the divergence in Figure 2. Further defects — unseeded randomness, unclipped predictions, quadratic-cost gradient accumulation, minibatch indices that revisit some samples while skipping others, and ranking metrics that count already-seen items as hits — are catalogued alongside the preserved code.

Four points of attribution and notation in the earlier text are also corrected in the present one: the 2007 paper is by Mnih and Salakhutdinov in that order [3]; BPMF [5] does not incorporate social relations or item content, an extension due to Liu *et al.* [1]; Equations (1) and (5) are likelihoods rather than posteriors; and the Frobenius norm applies to U and V rather than to individual factor vectors. One work cited there could not be located in any indexed venue, and the claim it supported is here supported by [2].

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